

# MAHARASHTRA METRO RAIL CORPORATION LIMITED

(A Joint Venture of Govt. of India & Govt. of Maharashtra)

## THANE INTEGRAL RING METRO RAIL PROJECT



## DESIGN BASIS REPORT FOR ELEVATED STATION

Revision - A

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MAHARASHTRA METRO RAIL CORPORATION LIMITED  
METRO BHAWAN, EAST HIGH COURT ROAD (VIP ROAD),  
OPPOSITE DR. BABASAHEB AMBEDKAR COLLEGE,  
NEAR DIKSHABHOOMI, NAGPUR - 440010, MH, India

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## **1. INTRODUCTION**

### **1.1 Brief Description of the Project**

The proposed Thane Metro corridor is about 29 Km long and have 22 stations, in which 20 stations are elevated and 2 stations are underground.

### **1.2 Scope**

The object of this Design Basis Document is to establish a common procedure for the design of “Elevated Stations for Thane Integral Ring Metro Rail Project”. This is meant to serve as guide to the designer but compliance with the rules there in does not relieve them in any way of their responsibility for the stability and soundness of the structure designed. The design of Elevated Stations requires an extensive and thorough knowledge and entrusted to only to specially qualified engineers with adequate practical experience in structural designs.

The DBR is only for structural design of Elevated Stations. Extended platform portion which is generally on single column or portal type structure shall be designed as part of viaduct.

The structural elements connected to the member on which metro live loads are supported may also be designed with taking loads applicable as specified in “Approved Design Basis Report (DBR) for Viaduct of Thane Ring Metro”. LWR forces shall be specified by the Metro, if RSI analysis is not practicable. Load combination as per “Approved Design Basis Report (DBR) for Viaduct of Thane Ring Metro” shall also be considered. Other structural elements such as secondary beams, stub columns etc., may be designed as per IS: 456.

Structures, where Metro Live loads are not applicable, the design of Plain and Reinforced Concrete structures will generally be governed by IS: 456, Pre-stressed Concrete structures shall generally be governed by IS: 1343. Steel structures design shall generally be governed by IS: 800. Seismic design shall be governed by IS: 1893.

### **1.3 Units**

The main units used for design will be: [t], [m], [mm], [kN], [kN/m<sup>2</sup>], [MPa], [°C], [rad].

## **2. DESIGN SPECIFICATION FOR STATION BUILDING**

### **2.1 Materials**

#### **2.1.1 Cement**

For Plain and Reinforced Concrete structures cement shall be used as per clause 5.1 of IS: 456 and in case of Pre-stressed Concrete structures as per clause 5.1 of IS: 1343.

#### **2.1.2 Concrete**

As per clause 6, 7, 8, 9 and 10 of IS: 456 in case of Plain and Reinforced Concrete structures and Clause 6, 7, 8, 9 and 10 of IS: 1343 for Pre-stressed Concrete structures.

Short term modulus of elasticity ( $E_c$ ) shall be taken as per clause 6.2.3.1 of IS: 456 for Plain and Reinforced Concrete structures and IS: 1343 for Pre-stressed Concrete structures.

The modular ratio for concrete grades shall be taken as per Annexure B of IS: 456.

The Density of concrete shall be as per IS: 456.

### **2.1.3 Pre-stressing Steel for Tendons**

As per clause 5.6.1 of IS: 1343.

#### **2.1.3.1 Young's Modulus**

As per pre-stressing steel used in accordance with Para 2.1.3 above.

#### **2.1.3.2 Pre-stressing Units**

As per clause 13 of IS: 1343.

#### **2.1.3.3 Maximum Initial Pre-stress**

As per clause 19.5.1 of IS: 1343.

#### **2.1.3.4 Density**

Weight of strands shall be as per relevant clauses of IS codes as per material being used as indicated in para 2.1.3 above.

#### **2.1.3.5 Sheathing**

As per clause 12.2 of IS: 1343.

### **2.1.4 Structural Steel**

Structural steel used shall confirm to

- Hollow steel sections as per IS: 4923.
- Steel for General Purposes as per IS: 2062.
- Steel tubes for structural purpose shall be as per IS: 1161.

Note:

- Grade of steel to be used shall be indicated, shall not be less than minimum grade as applicable, based on whether structure is taking moving loads or not and relevant code as indicated in note (ii) and (iii) below.
- Design of steel structure will be governed by IRS Steel Bridge Code in case structure is taking moving loads of Metro, otherwise will be governed by IS: 800. In case of composite (steel-concrete) structures it will be governed by IS: 11384 & IS: 3935.
- Fabrication shall be done in accordance with IRS BI (Fabrication Code) in case structure is taking moving loads of Metro, otherwise shall be done as per IS: 800.

### **2.1.5 Reinforcement**

As per clause 5.6 of IS: 456 for Plain and Reinforced Concrete structures and as per clause 5.6.2 of IS: 1343 for Pre-stressed Concrete structures.

Note: For Seismic zone III, IV & V, HYSD steel bars having minimum elongation of 14.5 percent and conforming to requirements of IS: 1786 shall be used.

#### **2.1.5.1 Reinforcement Detailing**

All reinforcement shall be detailed in accordance with clause 12 and 26 of IS: 456 for Plain and Reinforced Concrete structures, as per clause 12.3 and 19.6.3 of IS: 1343 for Pre-stressed Concrete structures. Ductile detailing of seismic resisting RC elements, shall comply with ductile requirements of IS: 13920.

### **2.2 Durability**

Durability of Concrete shall be as per clause 8.0 of IS: 456 for Plain and Reinforced Concrete structures, as per clause 8.0 of IS: 1343 for Pre-stressed Concrete structures and Section 15 of IS: 800 for Steel Structures.

#### **2.2.1 Concrete Grades**

The minimum grade of concrete for all structural elements including piles, blinding layers and levelling courses shall be as under:

| <b>Sr. No.</b> | <b>Structural component</b>                                                                                               | <b>Minimum Grade of Concrete</b> |
|----------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| A.             | Pre-tensioned girders                                                                                                     | M50                              |
| B.             | Superstructure - deck slab & beams,<br>Pier and Pier arm,<br>Portal beams,<br>Pedestal,<br>Shear key and Seismic stoppers | M40                              |
| C.             | Crash barrier, Pier protection                                                                                            | M40                              |
| D.             | Slabs, Beams, Walls, Columns                                                                                              | M35                              |
| E.             | Pile, Pile cap, Open foundation, Basement slab, Ancillary building foundation slab, Retaining wall                        | M35                              |
| F.             | Solid slab                                                                                                                | M40                              |
| G.             | Blinding concrete or levelling course                                                                                     | M15                              |

#### **2.2.2 Cover to Reinforcement**

As per clause 26.4 of IS: 456 for Plain and Reinforced Concrete Structures and clause 12.3.2 of IS: 1343 for Pre-stressed Concrete structures. Cover to pre-stressing steel shall be in accordance with clause 12.1.6 of IS: 1343.

#### **2.2.3 Fire Resistance Period**

All the structural elements in the station building shall be designed for a minimum fire resistance period of 2 hours. The minimum element thicknesses

for this fire resistance shall be as per clause 21 of IS: 456 for Concrete structures and as per section 16 of IS: 800 for Steel structures.

#### **2.2.4 Crack Width Check**

All structural concrete elements shall be designed to prevent excessive cracking due to flexure, early age thermal and shrinkage. Flexural crack width shall be checked in accordance with clause 35.3.2 and 43 of IS: 456 for Plain and Reinforced Concrete Structures and clause 20.3.2 and 24.2 of IS: 1343 for Pre-stressed Concrete structures.

### **2.3 Clearances**

- (i) Clearance for Road Traffic:** As per relevant IRC specifications and Road Authority requirements.
- (ii) Clearance for Railway Traffic:** Indian Railways Schedule of Dimensions (SOD) shall be applicable.
- (iii) Clearances for Metro Traffic:** As per approved SOD of MAHA-METRO.
- (iv) For utility services:** The clearances to utilities, drainage etc. shall be as mandated by the utility owner/ department.

### **2.4 Design Loads**

Elementary loads to be considered for design are: -

|                                          |      |
|------------------------------------------|------|
| Dead Loads                               | DL   |
| Super Imposed Loads                      | SIDL |
| Imposed (Crowd Live) Loads               | LL   |
| Earthquake Loads                         | EQ   |
| Wind Loads                               | WL   |
| Collusion/Impact Loads/Derailment Loads  | CL*  |
| Construction & Erection Loads            | EL   |
| Temperature Loads                        | OT   |
| Shrinkage                                | S    |
| Creep                                    | C    |
| Earth & Water Pressure                   | EP   |
| Surcharge Loads (Traffic, building etc.) | SR   |
| Pre-stress Force                         | PR   |
| Long Welded Rail Force                   | LWR  |
| Differential Settlement                  | DS   |

\*Load as applicable shall be taken.

#### 2.4.1 Dead Loads

Dead load shall be based on the actual cross section area and unit weights of materials and shall include weight of the materials that are structural components of Elevated Station and permanent in nature.

#### 2.4.2 Superimposed Dead Loads (SIDL)

Superimposed dead loads include all the weights of materials on the structure that are not structural elements but are permanent.

*Note: The SIDL can be of two types: Fixed or non-variable, and variable. In case Metro certifies that a portion of SIDL is of fixed or non-variable type and is not likely to vary significantly during the life of the structure and a special clause for ensuring the same is incorporated in the Metro's maintenance manual, the load factors applicable for dead load may be considered for this component of SIDL.*

The minimum distributed and concentrated loads shall be in accordance to IS: 875, wherever SIDL values are not available in relevant codes, the following values shall be adopted:

##### Stations

##### **For platform slab, the following assumptions will be taken:**

- Suspension load - 2.0 kN/m<sup>2</sup> uniform loads. (Suspension load will be considered as load of false ceiling, plumbing & electrical equipments, Escalator Pits etc. This load is applicable wherever necessary.)
- PSD - As per contractor's specifications.

##### **For the concourse area, the following assumption will be taken:**

- Suspension load - 2.0 kN/m<sup>2</sup> uniform loads. (Suspension load will be considered as load of false ceiling, plumbing & electrical equipments)
- Lift and Escalator support shall be designed as per manufacturer's details.

**NOTE:** The wall loads will be taken based on actual location shown in architectural drawings. External wall load/glazing load will be taken as per details provided in architectural drawings.

##### **SIDL for two tracks**

Details of SIDL for two tracks:

|                                         |           |
|-----------------------------------------|-----------|
| Cables                                  | 0.7 kN/m  |
| Cable troughs with cover                | 7.4 kN/m  |
| Cable trays                             | 0.1 kN/m  |
| Concrete plinths for rails + Rail + Pad | 31.0 kN/m |
| Miscellaneous (OCS, signaling,)         | 4.0 kN/m  |
| Hand Rail                               | 0.8 kN/m  |



Total: 44 kN/m for two tracks and 22 kN/m for one track.

Say 50 kN/m for two tracks and 25 kN/m for one track.

### 2.4.3 Imposed (Live) Loads

Imposed loads on station buildings are those arising from occupancy and the values includes, normal use by persons, furniture and moveable objects, vehicles, rare events such as concentrations of people and furniture, or the moving or stacking of objects during times of re-organisation of refurbishment, this shall be as per clause 19.3 of IS: 456.

#### **Live load on Platform, Concourse level, Staircases and Foot Bridge**

Live load should be considered as follows.

| Location                                                  | Distributed Load *                                       | Concentrated Load * |
|-----------------------------------------------------------|----------------------------------------------------------|---------------------|
|                                                           | kN/m <sup>2</sup>                                        | kN                  |
| <b>Public/Staff Room</b>                                  |                                                          |                     |
| Concourse Floor                                           | 5.0                                                      | 4.5                 |
| Staircase Area/Foot bridge                                | 5.0                                                      | 4.5                 |
| Platform                                                  | 5.0                                                      | 4.5                 |
| Office Accommodation                                      | 5.0                                                      | 4.5                 |
| Shop                                                      | 5.0                                                      | 4.5                 |
| <b>Technical Room</b>                                     |                                                          |                     |
| Heavy Equipment* Room<br>(ASS, UPS, Electrical<br>Switch) | 15.0 for ASS & Electrical<br>Switch **<br>25.0 for UPS** | 40.0                |
| Other Technical* Room                                     | 10.0**                                                   | 20.0                |

\* Please be checked for actual load and its location.

Floor slabs shall be designed for the distributed load as shown in the table here above. The floor slab also be checked, in critical locations, using the concentrated loads listed in the above table for possible local internal forces exceeding those produced by the normal distributed loads e.g. cantilevers, punching shear in slabs and all other locations sensitive to concentrated loads.

Concentrated loads and uniformly distributed loads are taken into account independently.

Dispersal of concentrated loads is at a spread-to-depth ratio of 1 horizontally to 1 vertically through concrete and 0.75 horizontally to 1 vertically for other surfacing.

\*\* Values are tentative & meant for preliminary design only. Actual loads will be calculated on the basis of equipment & machinery which have to be installed at detail design stage.

#### **2.4.4 Earthquake Loads**

Earthquake design shall follow the seismic requirements of IS: 1893 (Part-I). The provision as per “Approved Design Basis Report for Viaduct of Thane Ring Metro” shall be followed where structures are taking moving loads of metro.

##### **2.4.4.1 Drift Limitation**

The storey drift in the building shall satisfy the drift limitation specified in clause 7.11.1 in IS: 1893.

##### **2.4.4.2 Seismic Detailing**

- (i) For reinforced concrete structures as per IS: 13920.
- (ii) For other structures as per IS: 4326.

#### **2.4.5 Wind Loads**

The wind load shall be calculated as per IS: 875 part 3.

#### **2.4.6 Collision/Impact Loads/Derailment Loads**

- (i) For road traffic as per IRC 6.
- (ii) For metro as per IRS Bridge Rule.
- (iii) Clause 6.1.2 of IS: 875 (Part-5).

#### **2.4.7 Construction and Erection loads**

The weight of all temporary and permanent materials together with all other forces and effects which can operate from any part of structure during erection shall be taken into account. Allowances shall be made in the permanent design for any locked in stresses caused in any member during erection.

#### **2.4.8 Temperature**

As per clause 19.5 of IS: 456. Temperature gradient shall be considered as per clause 215 of IRC: 6, if applicable.

#### **2.4.9 Shrinkage**

The shrinkage strains shall be evaluated as per clause 6.2.4 of IS: 456 for Plain and Reinforced Concrete Structures and clause 6.2.4 of IS: 1343 for Pre-stressed Concrete structures.

For structure supporting Metro loading the effects of shrinkage as per clause 5.2.3 of IRS-CBC shall be considered.

#### **2.4.10 Creep**

The creep strains shall be evaluated as per clause 6.2.5 of IS: 456 for Plain and Reinforced Concrete Structures and clause 6.2.5 of IS: 1343 for Pre-stressed Concrete structures.

For structure supporting Metro loading the effects of creep as per clause 5.2.4 of IRS-CBC shall be considered.

#### **2.4.11 Earth & Water Pressure**

In the design of structures or parts of structures below ground level, such as retaining walls and underground pump room/water tank etc. the pressure

exerted by soil or water or both shall be duly accounted for. When a portion or whole of the soil is below the free water surface, the lateral earth pressure shall be evaluated for weight of soil diminished by buoyance and the full hydrostatic pressure. (As per IS: 875 Part 5).

All foundation slabs/footings subjected to water pressure shall be designed to resist a uniformly distributed uplift equal to the full hydrostatic pressure. Checking of overturning of foundation under submerged condition shall be done considering buoyant weight of foundation.

If any of the structure supporting Metro Loading is subjected to earth pressure, the loads and effects shall be calculated in accordance with clause 5.7 of IRS Substructure Code.

#### **2.4.12 Surcharge Load**

In the design of structures or parts of structures below ground level, such as retaining wall and underground pump room/water tank etc. the pressure exerted by surcharge from stationary or moving load, shall be duly accounted for.

#### **2.4.13 Pre-stressing Force (PR)**

The pre-stressing Force should be as per IS: 1343.

#### **2.4.14 Long Welded Rail Force**

Guidelines vide BS report No.119 “RDSO Guidelines for carrying out Rail-Structure interaction studies on Metro (Version-2)” shall be followed.

A rail structure interaction [RSI] analysis is required because the continuously welded running rails are continuous over the deck expansion joints. The interaction occurs because the rails are directly connected to the decks by fastening system.

1. Rail structure interaction studies shall be done as per provisions of UIC 774-3 R with the following parameters specified in consultation with track design engineers:
  - i) Track resistance in loaded and unloaded conditions.
  - ii) Maximum additional stresses in rail in tension as well as compression on account of rail-structure interaction.
  - iii) Maximum vertical deflection of the girder at ends.
2. Software and general methodology to be used for carrying out Rail-Structure interaction analysis must be validated before adopting the same.
3. Representative stretches must be chosen for carrying out Rail-Structure interaction.
4. Checks must be performed for break in rail continuity due to unusual conditions fractures or for maintenance purposes.
5. RDSO Guidelines for carrying out RSI studies shall be preferred.

6. LWR forces shall be considered in appropriate load combination as per IRS-CBC.

#### **2.4.15 Settlement**

Maximum and differential settlement shall not exceed, as provided in Table 1 of IS: 1904.

#### **2.4.16 Other Forces and Effects**

As per clause 19.6 of IS: 456.

### **2.5 Design Load Combination**

#### **2.5.1 Ultimate Load Combinations**

Each component of the structure shall be designed and checked for all possible combinations of applied loads and forces. They shall resist effect of the worst combination. Following shall be considered: -

- (i) Load combinations and factors as per Table 18 of IS: 456 Plain and Reinforced Concrete structures.
- (ii) Load combination and factors as per Table 7 of IS: 1343 for Pre-stressed Concrete structures.
- (iii) Load combination as per section 3 and factors as per section 5 of IS: 800 for Steel structures.
- (iv) Load combination as per clause 6.3 of IS: 1893 (Part-I).
- (v) Load combinations as per IRS-CBC and RDSO guidelines for Seismic design of Railway Bridges where Metro live loads are applicable.

Note:

- (i) Load combination for construction load case shall be decided by Metro as per methodology of construction.
- (ii) Reference of IRC: 6 or IS: 875 (Part5) be taken for collision case if collision of road vehicles is involved.

#### **2.5.2 Serviceability Load Combinations**

The following load combinations and load factors shall be used for design for serviceability limit state:

- (i) Load combinations and factors as per Table 18 of IS: 456 for Plain and Reinforced Concrete structures.
- (ii) Load combination and factors as per Table 7 of IS: 1343 for Pre-stressed Concrete structures.
- (iii) Load combination as per section 3 and factors as per section 5 of IS: 800 for Steel structures.
- (iv) Load combinations as per IRS-CBC where Metro live loads are applicable.

## **2.6 Deflection Criteria**

The deflection limitations as per clause 23.2 of IS: 456 for Plain and Reinforced Concrete structures and clause 20.3.1 of IS: 1343 for Pre-stressed Concrete structures shall be followed.

### **2.6.1 Lateral Sway**

The lateral sway at the top of the building due to Wind loads should not exceed  $H/500$ , where H is the height of the building.

## **2.7 Fatigue Check**

Fatigue phenomenon needs to be analysed only for those structural elements that are subjected to repetition of significant stress variation (under traffic load). Fatigue checks for

**(i) RCC and PSC structures** – As per clause 13.4 of IRS-CBC.

**(ii) Steel Structures** –

- (a) In case of Metro live loads, as per clause 3.6 of IRS Steel Bridge Code shall govern. If  $\lambda^*$  values are required to be used, the train closest to the actual train formation proposed to be run on the metro system shall be used. Otherwise, detailed counting of cycles shall be done.
- (b) For other cases as per Section 13 of IS: 800.

\*Damage equivalence factors (As per IRS Steel Bridge Code).

## **2.8 Foundations**

### **2.8.1 Types of Foundation**

Considering the nature of ground; type of proposed structures, expected loads on foundations, the following type of foundations are considered practical:

- (a) Spread or pad footing
- (b) Raft foundation
- (c) Pile foundation

No matter the type of foundation to be adopted, the following performance criteria shall be satisfied:

- 1) Foundation must not fail in shear
- 2) Foundation must not settle by more than the settlements permitted as per Table-1 of IS: 1904.

### **2.8.2 Design of Pile**

IS: 2911 shall be followed for design of pile, load capacity etc.

#### **Pile Settlement**

Methods of estimating the settlement of deep foundations depend upon the type of deep foundation and the manner of transfer of loads from the structure to the soil. Theoretical estimation of settlement shall be done in accordance with IS: 8009 (Part II) by integrating the vertical strain for the entire depth of soil and rock formation.

The settlement of each pile and/or pile group should be determined and it should be demonstrated that such total and/or differential settlement can be tolerated by the structure.

### **2.8.3 Foundations**

IS: 1904 shall be followed for design of foundations in soil. The safe bearing capacity for shallow foundations shall be calculated in accordance with IS: 6403.

#### **Computation of Settlements of Foundations**

The calculation for settlement of foundations shall be done as per:

- IS: 8009 Part-1 for shallow foundations
- IS: 8009 Part-2 for deep foundations

### **2.9 Design of Water Retaining Structure**

It should be designed as per IS: 3370.

### **3. List of Design Codes and Standards**

The designs of station buildings shall be carried out as per provisions of this Design Specifications. Reference shall be made to following codes for any additional information.

Order of preferences of codes shall be as follows: -

- (i) IS
- (ii) IRS
- (iii) IRC
- (iv) BS or Euro Code
- (v) AASHTO

(**Note:** Each time latest code with the latest correction slip shall be adopted)